



Impact of volatile fatty acids on microbial electrolysis cell performance



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HIGHLIGHTS

- Acetate can be readily oxidized by ARB.
- Propionate is a less favorable substrate than acetate and butyrate in MEC.
- Butyrate and propionate are not consumed directly by ARB.
- The ratio of suspended biomass to attached biomass in MEC is approximately 1:4.

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ABSTRACT

This study investigated the performance of microbial electrolysis cells (MECs) fed with three common fermentation products: acetate, butyrate, and propionate. Each substrate was fed to the reactor for three consecutive batch cycles. The results showed high current densities for acetate, but low current densities for butyrate and propionate (maximum values were 6.0 ± 0.28 , 2.5 ± 0.06 , 1.6 ± 0.14 A/m², respectively). Acetate also showed a higher coulombic efficiency of $87 \pm 5.7\%$ compared to 72 ± 2.0 and $51 \pm 6.4\%$ for butyrate and propionate, respectively. This paper also revealed that acetate could be easily oxidized by anode respiring bacteria in MEC, while butyrate and propionate could not be oxidized to the same degree. The utilization rate of the substrates in MEC followed the order: acetate > butyrate > propionate. The ratio of suspended biomass to attached biomass was approximately 1:4 for all the three substrates.

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1. Introduction

Hydrogen plays a key role in sustainable energy production. Although hydrogen can be recovered by fermentation of organic material rich in carbohydrates, the majority of organic matter remains in the form of volatile fatty acids (VFAs). The primary fermentation end products during biohydrogen production are acetic, butyric, and propionic acids (Liu et al., 2005a). To achieve a higher conversion of a substrate to hydrogen, an alternative to fermentation for hydrogen production is the process of electrohydrogenesis using microbial electrolysis cells (MECs). Anode-respiring bacteria (ARB), such as *Geobacter Shewanella*, *Clostridium*, *Pseudomonas*, *Desulfuromonas*, *Eseherichia*, and *Klebsiella*, are able to transmit their electrons to a solid electron acceptor as part of their energy-generating respiration (Lee et al., 2010; Torres et al., 2007, 2010). Three mechanisms of extracellular electron transfer

have been proposed, i.e., direct electron transfer, electron shuttle, and via a solid conductive matrix (Torres et al., 2010). The energy in the electrons can be utilized for electricity generation in a microbial fuel cell (MFC) (Logan et al., 2006) or for hydrogen gas production in a microbial electrolysis cell (MEC) (Liu et al., 2005b). In MECs, ARB are of special interest for oxidizing biodegradable organic compounds present in wastes and other forms of biomass into protons, electrons, and bicarbonate (Lee et al., 2010; Torres et al., 2007). The electrons reach the cathode and react with water to produce hydrogen. Hydrogen production using MECs has been studied using simple organic compounds, such as acetate, propionate, glucose, glycerol (Cheng and Logan, 2007; Lu et al., 2012; Selembo et al., 2009; Sun et al., 2010); complex organic matter, such as starch, protein (Montpart et al., 2015; Nam et al., 2014); and real wastewater, for example, domestic wastewater, winery wastewater, and industrial wastewater (Cusick et al., 2011; Ditzig et al., 2007; Tenca et al., 2013).

Recently, combining dark fermentation with MECs seems to be very promising. Anode respiration process and fermentation can be combined in two different ways. One is adding the fermentative

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microorganisms and anode respiring bacteria in the same reactor to create a mixture of these two cultures in the MEC anode chamber. Montpart et al. (2015) obtained a group of microorganisms able to degrade a specific complex substrate (glycerol, milk and starch) by separately growing fermentative and ARB microbial communities in culture flasks and in an MFC respectively before combining both communities in a single chamber MEC. In this approach, they demonstrated that the growth of an anodic syntrophic consortium between fermentative bacteria and ARB was operationally enhanced and increased the potential of these complex substrates to be treated (Montpart et al., 2015). On the other hand, fermentation and hydrolysis could be separated into an independent reactor, with the MFC/MEC receiving simpler organic compounds typical of fermentation effluent, which are further consumed by ARB (Torres et al., 2007). For example, a two-stage dark-fermentation and electrohydrogenesis process was used to produce hydrogen gas by converting organic compounds such as cellulose (Lalaurette et al., 2009) and crude glycerol (Chookaew et al., 2014) to smaller compounds.

As the main end products from dark fermentation, VFAs have a vital impact on the performance of MECs. Escapa et al. (2013) found that acetate and butyrate were easily degradable, whereas propionate exhibited pseudo-recalcitrant behavior in a continuous two-chamber MEC fed with synthetic dark fermentation wastewater. However, this was contradictory to the findings of other groups. Li et al. (2014) indicated that the propionate had a higher priority sequence for hydrogen production than butyrate in a single-chamber MEC fed with corn stalk fermentation effluent. In their work, the removal efficiency of acetate, propionate and butyrate were reported as of 81–91%, 11–16% and 4%, respectively (Li et al., 2014). Torres et al. (2007) also demonstrated that acetate and propionate were consumed more effectively than the butyrate in the continuous-flow H-type MEC fed with a mixture of fermentation products. They reported a maximum current density for acetate of 9.0 A/m², 1.6 A/m² for propionate, and only 0.16 A/m² for butyrate. The detailed comparisons among the above studies are listed in Table 1. In order to clear this contradiction and figure out the impact of different VFAs on the MEC performance, this study compared MEC operational parameters by feeding the MEC with different VFAs, namely acetate, butyrate and propionate.

2. Methods

2.1. Reactor set-up

The MEC was fabricated from plexiglass with anode and cathode volumes of 550 mL and 225 mL, respectively. The liquid volume in the anode varied from 500 mL to 530 mL since some of

the liquid was washed out with the purge of nitrogen. One bundle of high density carbon fibers (2293-B, 24 K Carbon Tow, Fibre Glast Developments Corp., OH, USA) that was intertwined through holes drilled on a stainless steel frame was used as the anode module. The specific surface area of the fibers was 571,429 m²/m³ (fiber's diameter, 7 μm; length, 150 cm). The bundle contained 24,000 individual carbon filaments with a geometric surface area of 7913 cm². The geometric surface area of the anodes per MEC anode volume was 1583 m²/m³. The carbon fibers were pretreated with nitric acid (1 N), acetone (1 N) and ethanol (1 N) for 1 day each, and then washed with MilliQ water (18.2 MΩ-cm) (Dhar et al., 2013). The cathode electrode was made of a stainless steel mesh (Type 304, McMaster Carr, OH, USA). An anion exchange membrane (AMI-7001, Membrane International Inc., NJ, USA) was placed between the anode and the cathode as a separator, and the geometric surface area of the membrane was 18 cm². The membrane was pretreated at 40 °C in 5% NaCl solution for 24 h as per the manufacturer recommendations. To avoid possible short-circuiting and liquid leakage, non-conductive polyethylene mats were used between the electrodes and membrane (Dhar et al., 2013). The distance between the anode and cathode electrodes was less than 1 cm. A schematic and picture of the sandwich type anode-membrane-cathode are shown in Fig. 1.

A voltage of 1.0 V was applied across the electrodes using a power supply (B&K Precision Corp., California, USA). The positive lead of the power supply was connected to the anode, and the negative lead was serially connected to a 10 Ω resistor and the cathode. The temperature was maintained at room temperature (25 °C) during the whole experiment.

2.2. MEC inoculation and operation

The MEC was inoculated with 50 mL of effluent from a working MEC, which was selectively enriched from an activated sludge microbial consortium (Dhar et al., 2013). The anode chamber was fed with a medium containing 2.3 g/L KH₂PO₄, 4.66 g/L Na₂HPO₄, 0.038 g/L NH₄Cl and 0.84 g/L NaHCO₃ and 1 mL/L of a trace element mixture with the following composition: 25 mg/L MgCl₂·6H₂O; 6 mg/L MnCl₂·4H₂O; 1.2 mg/L CaCl₂·2H₂O; 0.5 mg/L ZnCl₂; 0.11 mg/L NiCl₂; 0.1 mg/L CuSO₄·5H₂O; 0.1 mg/L AlK(SO₄)₂·12H₂O; 1 mg/L Co(NO₃)₂·6H₂O; 0.1 mg/L H₃BO₃; 5 mg/L EDTA; 0.1 mg/L Na₂WO₄·2H₂O; 0.1 mg/L NaHSeO₃; 0.2 mg/L Na₂MoO₄·2H₂O. 20 mM FeCl₂·4H₂O and 77 mM Na₂S·9H₂O were also added to the medium (1 mL/L) (Dhar et al., 2013; Torres et al., 2007). The substrate concentrations (added as sodium acetate, sodium propionate, sodium butyrate) are noted below. Medium pH was constant at 7.2 ± 0.2. The cathode chamber was filled with distilled water.

Table 1
Summary of current densities and removal efficiencies by different authors using acetate, butyrate and propionate.

Substrate	Running mode	MEC type	Applied voltage (V) (unless otherwise stated)	T ^a (°C)	pH	Influent component	Influent COD (mg/L)	Removal efficiency (%)	CD ^d (A/m ²)	HPR ^c (m ³ /m ³ /d)	Source
CSFE*	batch	Single chamber	0.8	36	7	acetate butyrate propionate	1490 ^e 1967 ^e 45 ^e	91 4 14 ^d	340	3.43	Li et al. (2014)
SDFE [§]	continuous	Two chamber (gas cathode)	1	25	7	acetate butyrate propionate	1302 ^e 736 ^e 1227 ^e	100 100 <100	206	1.42	Escapa et al. (2013)
SDFE [§]	continuous	H-type dual-compartment	Anode potential (+0.1 V vs Ag/AgCl)	30	7.4	acetate butyrate propionate	2560 ^e 6400 ^e 4480 ^e		281 ^e 5 ^e 50 ^e		Torres et al. (2007)

Note: T^a: temperature; CD^b current density; HPR^c: hydrogen production rate; ^d: the data was gotten from the figure in the reference; ^e: the data was calculated based on the information in the literature; CSFE *: corn stalk fermentation effluent; SDFE [§]: synthetic dark fermentation effluent.

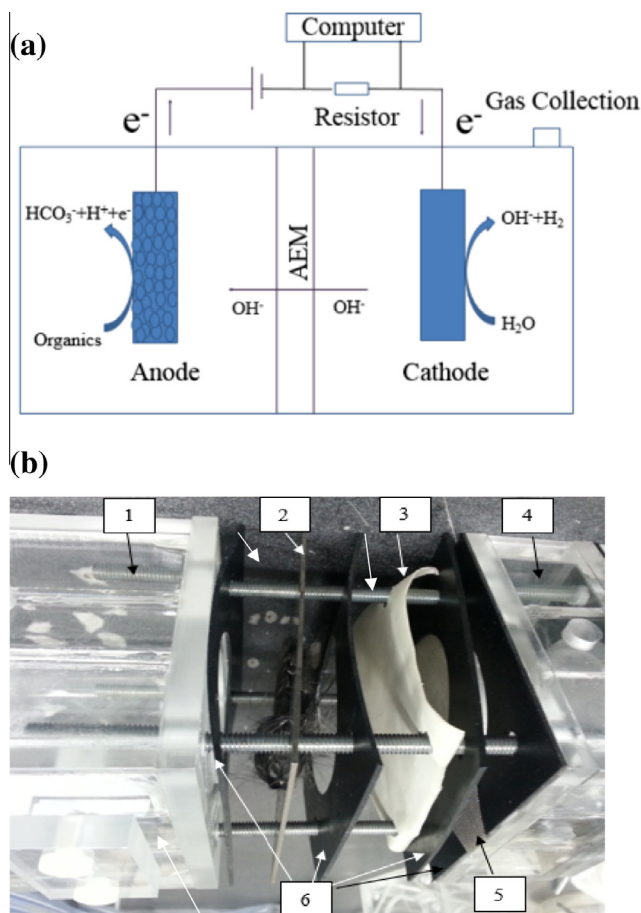


Fig. 1. (a) Schematic illustration of a typical two-chamber MEC with an anion exchange membrane (AEM); (b) Picture of connecting the anode chamber (1), anode electrode (2), membrane (3), cathode chamber (4), cathode electrode (5), and non-conductive polyethylene (6) together.

The MEC was carried out in batch mode. At least three consecutive batch cycles were achieved before changing the substrate. When the current dropped below 2 mA for acetate and butyrate-fed cycles, and 1 mA for propionate-fed cycles, the liquids in the anode and cathode chamber were emptied and refilled with the medium as described above. 2 g/L sodium acetate (CH_3COONa) corresponding to a COD of 1600 mg/L was added during the start-up period and three consecutive cycles. Subsequently, 0.55 g/L sodium butyrate ($\text{C}_4\text{H}_7\text{COONa}$) were fed into the MEC. Finally, 0.686 g/L of sodium propionate ($\text{C}_3\text{H}_5\text{COONa}$) was added and the MEC was run for another three cycles. To reduce the cycle time, the influent COD of butyrate and propionate were both reduced to 800 mg/L. Ultra-pure nitrogen was sparged into the anode chamber for 20 min at the beginning of each batch cycle to ensure anaerobic conditions.

2.3. Analytical methods

The total volume of biogas produced from the MECs was measured using the water displacement method. The biogas composition including hydrogen, methane, and nitrogen was determined by a gas chromatograph (Model 310, SRI Instruments, Torrance, CA) equipped with thermal conductivity detector (TCD) and a molecular sieve column (Mole sieve 5A, mesh 80/100, 6 ft × 1/8 in) (Gupta et al., 2014). Argon was used as a carrier gas at a flow rate of 30 mL/min and the temperature of the column and thermal conductivity detector (TCD) detector were 90 °C and 105 °C,

respectively. In the MEC, the voltage drop across the external resistor was measured using a multimeter with a data acquisition system (2700, Keithly Instruments Inc., Cleveland, Ohio), with the current calculated using Ohm's Law ($I = V/R$), where V was the measured voltage drop across the resistor ($R = 10 \Omega$) (Rader and Logan, 2010). Total and soluble chemical oxygen demand (TCOD/SCOD) were measured using HACH methods and test kits (HACH Odyssey DR/2500 spectrophotometer manual). TSS and VSS were analyzed using standard method (APHA, 1998). pH was measured using a pH probe (SympHony B10P, VWR, Visalia, CA).

2.4. Calculations

2.4.1. Hydrogen recovery

Coulombic efficiency (C_E) was calculated on the basis of the measured current compared to the substrate removed using the following Eqs. (1–3):

$$C_E = \frac{n_{CE}}{n_{th}} = \frac{8 \cdot I_{AVG,90} \cdot t}{F \Delta \text{SCOD}} \quad (1)$$

$$n_{th} = \frac{F \Delta \text{SCOD}}{M_{O_2}} \quad (2)$$

$$n_{CE} = \frac{I_{AVG,90} t}{2F} \quad (3)$$

where n_{CE} (mol) is the moles of hydrogen that could be recovered based on measured current, n_{th} (mol) is the maximum theoretical hydrogen potential from the SCOD removal, F is the Faraday constant ($F = 96,485 \text{ C/mole}^-$), ΔSCOD (g) is the soluble COD removed, and 8 is the conversion factor of COD to moles of electrons, M_{O_2} (g/mol) is the molecular weight of oxygen, while the 2 in Eq. (2) is the number of moles of hydrogen that can be produced with each mole of SCOD consumed. $I_{AVG,90}$ is the average current calculated over the time (t) for accumulation of 90% of the charge. The use of $I_{AVG,90}$ is more accurate when analyzing MEC performance in a batch cycle because it eliminates the small current densities at the end of the cycle and focuses on the most useful part of the current generation cycle (Ivanov et al., 2013).

The cathodic hydrogen recovery (r_{cat}) was calculated using Eq. (4):

$$r_{cat} = \frac{n_{H_2}}{n_{CE}} \quad (4)$$

where n_{H_2} (mol) is the actual moles of hydrogen recovered at the cathode.

The overall hydrogen recovery is (Logan et al., 2008):

$$R_{H_2} = C_E r_{cat} \quad (5)$$

Theoretical hydrogen yield (Y_{H_2}) is based on the theoretical maximum production of hydrogen:

$$Y_{H_2} = \frac{n_{H_2}}{n_{th}} \quad (6)$$

where n_{th} is the maximum theoretical hydrogen (mol) based on SCOD removal.

Volumetric hydrogen production rate (HPR) ($\text{m}^3 \text{H}_2/\text{m}^3/\text{d}$) was normalized to the cathode liquid volume (225 mL).

2.4.2. Energy recovery

The energy recovered (η_E) based on the energy input was calculated using the following equation:

$$\eta_E = \frac{-W_{H_2}}{W_{in}} = \frac{n_{H_2} \Delta H_{H_2}}{E_{ap} I t - I^2 R t} \quad (7)$$

where W_{H_2} (J) is the amount of energy recovered in hydrogen, W_{in} (J) is the electrical energy input, ΔH_{H_2} (-285.8 J/mol) is the energy content of hydrogen based on the heat of combustion (Logan et al., 2008), E_{ap} (V) is the applied voltage to the system by the power supply, I (A) is the current during the batch cycle, R is the external resistor (10Ω), and t (s) is the time of each batch cycle.

The energy recovered (η_{E+S}) based on both the energy input and the energy in the substrate was calculated using the following equation:

$$\eta_{E+S} = \frac{-W_{H_2}}{W_{in} - W_s} \quad (8)$$

where W_s is the energy in the substrate which can be calculated similar to the energy content of hydrogen, i.e., $\Delta H_{Acetate} = -874.3$ kJ/mol (Logan et al., 2008), $\Delta H_{Propionate} = -1528.3$ kJ/mol (Chadwick, 1988), $\Delta H_{Butyrate} = -2183.5$ kJ/mol (Dorofeeva et al., 2001).

The current density (CD) (A/m^2 or A/m^3) was the current produced in the batch cycle per unit membrane surface area, or unit liquid volume.

3. Results and discussion

3.1. Effects of substrate on current density and hydrogen production rate

The profile of the batch current density fed with different substrates is illustrated in Fig. 2. During the start-up period, the current stayed at zero for 24 h, and then started to increase. For all other cycles, the current would increase directly (without lag phase) after feeding. This indicated that the anode respiring bacteria were effectively attached to the anode. When the MEC was fed with butyrate for the first time (cycle 5), the current increased smoothly peaking on day 51, 9 days after the butyrate feed, thus demonstrating that the ARB were adapting to the new substrate. In the last butyrate cycle, the current peaked only 24 h after the feed. For all three butyrate batches, the maximum current was almost the same at 4.5 mA. After feeding the MEC with propionate, the current also increased very slowly and after 8 days achieved a lower maximum current of 3.0 mA. In the last cycle of the propionate-fed MEC (cycle 10), the current did not peak until after 8 days, which was the same as the first propionate cycle (cycle 8). The comparatively slower rate for the current to peak in the propionate-fed MEC denoted that the ARB cannot utilize propionate directly. It is more likely that the propionate was first oxidized to acetate by acetogenic bacteria, and then consumed by ARB. The average time for accumulation of 90% of the charge were

8.9 days for acetate, 9.1 for butyrate and 12 days for propionate. The longer time for propionate to accumulate 90% of the charge was mainly due to the slower rate for the current density to peak.

The cycles fed with acetate had the highest peak of current density ($6.0 \pm 0.28 A/m^2$), followed by the butyrate fed cycles ($2.5 \pm 0.06 A/m^2$), and propionate fed cycles achieved the lowest current density ($1.6 \pm 0.14 A/m^2$). The differences between the current densities might be attributed to a number of factors including the substrates order, the resistance of the membrane, pH gradient between the anode and cathode chambers, the influent COD (COD_{in}), and the substrate utilization rate. Of the aforementioned factors, the order of the feeding and the resistance of the membrane were discounted because at the end of this experiment, the MEC was fed with acetate again at a concentration of 800 mg COD/L, and the current was almost the same as the current achieved at the beginning of this experiment (See Fig. 2). The pH gradient was not a reason to cause these differences neither, since the pHs of the cathode were very close, i.e., 9.8 ± 1.7 for acetate, 9.7 ± 0.8 for butyrate, and 10.0 ± 0.4 for propionate. The anode pHs during the whole experimental period were maintained near neutral by the buffer in the medium. Even though the influent COD of acetate was different from that of butyrate and propionate, this could not cause the large difference in the maximum current densities between acetate and butyrate or propionate, because at the end of the experiment, when fed with 800 mg/L COD of acetate, the achieved current density was as high as $5.5 A/m^2$, which was almost the same as MEC feeding with 1600 mg/L COD acetate. The reduced influent COD concentration only reduced the cycle time from 10.6 to 6.4 days. This is consistent with the findings of Liu et al. (2005a) who reported that the voltages generated in a microbial fuel cell (MFC) using acetate at different concentrations (from 80 mg/L to 800 mg/L) stayed at around 0.45 V. In addition, Oh and Logan (2005) noted that the current density remained the same upon changing the influent propionate concentration from 0.26 mM to 0.53 mM. This clearly demonstrates that the only reason to limit the current densities in this case was the type of substrate. Acetate has been well known to be easily utilized by ARB, since it is not fermentable and has relatively rapid oxidation kinetics in MFC/MECs (Lee et al., 2009), while there has been no agreement on the relative biodegradability of butyrate and propionate. As the substrate-utilization rate is proportional to current density in an MEC (Lee and Rittmann, 2009), the utilization rate of the propionate (28 ± 0.8 mg COD/L/d) might have limited the current density, which was relatively low when compared with the average substrate utilization rate of acetate (92 ± 5.0 mg COD/L/d) and butyrate (41 ± 4.8 mg COD/L/d). This indicated that the utilization rate of the substrates for ARB in MECs followed the order: acetate > butyrate > propionate.

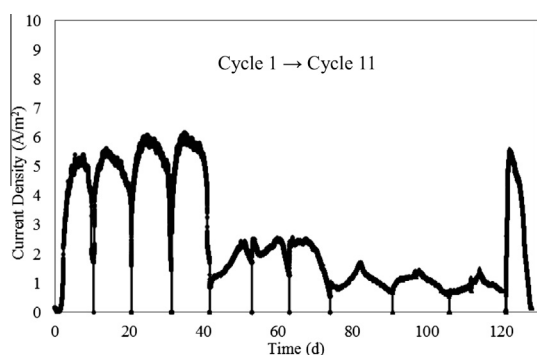


Fig. 2. The changes of current densities with different substrates, including: start-up cycle (1), acetate-fed cycles (2–4), butyrate-fed cycles (5–7), propionate-fed cycles (8–10), and acetate-fed cycle (11). The COD of cycle 1–4 are 1600 mg/L, and for cycle 5–11 are 800 mg/L.

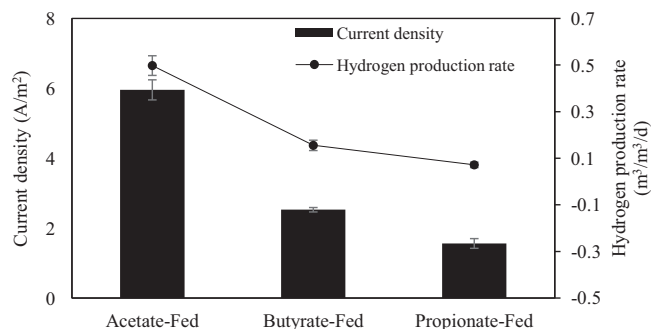


Fig. 3. The average current densities and hydrogen production rates in MEC fed with different substrates.

As depicted in Fig. 3, the type of substrate also has a significant impact on the hydrogen production rate. Similar to the trend of current density, the hydrogen production rate decreased from $0.50 \pm 0.04 \text{ m}^3/\text{m}^3/\text{d}$ for acetate to $0.07 \pm 0.01 \text{ m}^3/\text{m}^3/\text{d}$ for propionate. The current density and hydrogen production exhibited the same trend.

3.2. Effects of substrate on hydrogen recovery and energy efficiency

Cathodic hydrogen recovery is a very important parameter to evaluate the MEC performance, since it takes into account the H_2 recovered at the cathode and the electrons transferred through the electrode. While indeed both electrodes appear independent, based on Eq. (4), the cathodic hydrogen recovery depends on both the hydrogen recovered and the electrons transferred from the anode to the cathode, which is influenced by the ARB activity, and hence varied from one substrate to another. According to the cathodic reaction, the more the electrons transferred to hydrogen, the higher the cathodic hydrogen recovery. In this study, the cathodic hydrogen recovery decreased in the order of feeding with acetate, butyrate and propionate as shown in Fig. 4, which were 98 ± 0.8 , 79 ± 4.9 , and $71 \pm 7.2\%$, respectively. The differences in the cathodic hydrogen recoveries revealed that the electrons transferred to produce hydrogen for propionate was not as efficient as butyrate and acetate. The ratio of current density over hydrogen production rate exhibited an inverse linear relationship as shown in Fig. 5.

The coulombic efficiencies for acetate, butyrate, and propionate were 87 ± 5.7 , 72 ± 2.0 and $51 \pm 6.4\%$. These values are comparable with previous studies in continuous-flow mode MECs of 86% for acetate (Torres et al., 2007), 41% for propionate (Torres et al., 2007); 23.6 $\pm$ 9.6% from dark fermentation effluent reported by Chookaew et al. (2014). The overall H_2 recovery is the product of coulombic efficiency and cathodic hydrogen recovery from Eq. (5), so the overall H_2 recovery exhibited the same trend as both the coulombic efficiency and cathodic hydrogen recovery.

The hydrogen yield and energy efficiency are shown in Fig. 6. Theoretically, 4 mol of hydrogen are produced with 1 mol of acetate consumed according to Eq. (9). In this study, 3.4 ± 0.25 mol hydrogen per mole of acetate consumed were observed. This hydrogen yield corresponded to an energy recovery of $161 \pm 1.4\%$ when evaluated in terms of only the voltage addition (1 V). For butyrate, the achieved hydrogen yield was 5.6 ± 0.29 mol H_2 /mol butyrate, as compared with a theoretical value of 10 mol H_2 /mol butyrate (Eq. (10)). The energy efficiency for butyrate was $121 \pm 7.3\%$. These data are comparable with the reported performance in the literature (Table 2). In this study, a hydrogen yield of 2.46 ± 0.17 mol H_2 /mol propionate was achieved as compared with a theoretical yield of 7 mol H_2 /mol propionate (Eq. (11)).

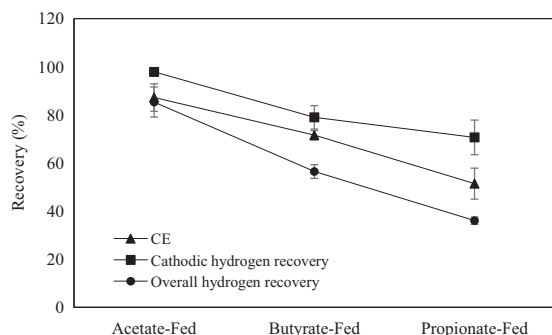


Fig. 4. The coulombic efficiencies (CE), cathodic hydrogen recoveries and overall hydrogen recoveries in MEC fed with different substrates.

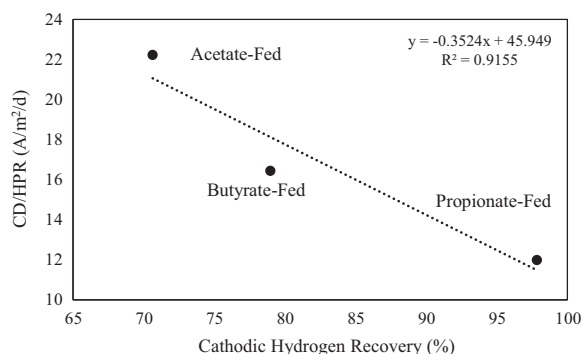


Fig. 5. The relationship of CD/HPR (current density/hydrogen production rate) and cathodic hydrogen recoveries in MEC fed with different substrates.

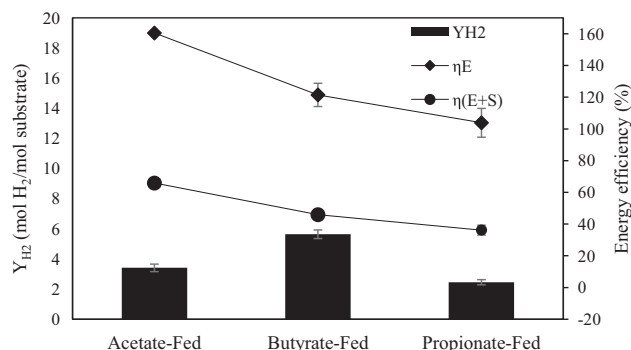
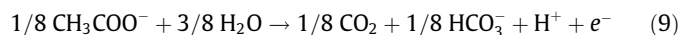


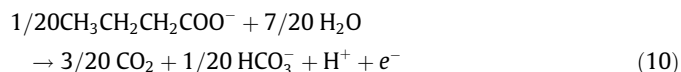
Fig. 6. The changes of hydrogen yields (Y_{H_2}), energy efficiencies with only electric input (η_E), and energy efficiencies including both electric input and the energy content in substrate (η_{E+S}) in MEC fed with different substrates.

The relatively lower observed hydrogen yield for propionate in this study is consistent with the literature. Moreover, the normalized hydrogen production per unit mass of soluble COD consumed (ΔSCOD) are 0.053 mol H_2 /g ΔSCOD for acetate, 0.035 mol H_2 /g ΔSCOD for butyrate and 0.022 mol H_2 /g ΔSCOD for propionate, confirming that more soluble COD was oxidized to produce hydrogen from acetate and butyrate-fed cycles than from propionate-fed cycles.

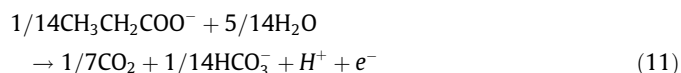
Acetate as substrate:



Butyrate as substrate:



Propionate as substrate:



3.3. Effects of substrate on COD removal and biomass production

Tables 3 and 4 describe the COD removal efficiencies together with a COD mass distribution for acetate, butyrate, and propionate. The start-up period was not included in this analysis. Almost all the SCOD was consumed ($96 \pm 1.7\%$) in the acetate-fed MEC. This was consistent with the high hydrogen yield obtained in this study.

Table 2

Comparison of key parameters reported in literatures versus data obtained in this study.

Substrate	E_{ap} (V)	Y_{H_2} (mol H_2 /mol substrate) (unless otherwise stated)	R_{H_2} (%)	HPR ($m^3/m^3/d$)	η_E (%)	η_{E+S} (%)	CD ^a (A/m^3)	CD ^b (A/m^2)	C_E (%)	Source
Acetic acid	0.8	3.65	91	1.1	260	82	99			Cheng and Logan (2007)
	0.5		53	0.02	169	53	2.8		92	Rozendal et al. (2006)
Sodium acetate	0.6		62	0.53	204	58	52		75	Hu et al. (2008)
	1		24	0.31			26		23	Rozendal et al. (2007)
	1	3.9		6.9*				4.7		Tartakovsky et al. (2009)
	1	1.4		1.33*				1.8		Tartakovsky et al. (2009)
	1	3.6	90	0.53	161	68	22.17	6.16	91	This study
Sodium butyrate	1	5.94	59	0.18	127	48	9.27	2.57	70	This study
Sodium propionate	1	2.56	37	0.072	112	43	5.26	1.46	59	This study
DFE [§]	1	27.93 mL H_2 /g COD consumed	1.9	0.019			23.5		9.1	Chookaew et al. (2014)

Note: The data was the highest value chosen from each substrate-fed cycles. E_{ap} : applied voltage; Y_{H_2} : the hydrogen yield based on substrate consumed; R_{H_2} : overall hydrogen recovery; HPR: average hydrogen production rate; η_E : energy efficiency based on electric energy input; η_{E+S} : energy efficiency based on both energy input and the energy content in substrate; CD^a: current density based on anode liquid volume; CD^b: current density based on anode or membrane surface area; C_E : coulombic efficiency; *: the data was calculated based on the information in the literature; §: dark fermentation effluent.

Table 3

COD data for each cycle.

Substrate	TCOD initial (mg)	SCOD initial (mg)	TCOD final (mg)	SCOD final (mg)	SCOD removed efficiency (%)	Average SCOD removal efficiency (%)
acetate-fed	894.4	894.4	82.3	39.8	95.6	96 ± 1.7
	895.6	895.6	48.2	22.4	97.5	
	797.7	797.7	83.5	46.6	94.2	
Butyrate-fed	420.1	420.1	102.6	74.2	82.3	88 ± 5.3
	408.1	408.1	58.8	39.8	90.3	
	424.9	424.9	89.0	32.9	92.3	
Propionate-fed	401.9	401.9	70.7	26.8	93.3	87 ± 5.8
	419.0	419.0	129.2	72.6	82.7	
	397.8	397.8	109.1	63.6	84.0	

Table 4

COD mass distribution.

COD sinks	Acetate-fed MEC		Butyrate-fed MEC		Propionate-fed MEC	
	COD (mg)	Fraction (%)	COD (mg)	Fraction (%)	COD (mg)	Fraction (%)
Initial COD	863	100	418	100	406	100
Final SCOD	36	4	49	12	55	14
H_2	655	76	208	50	124	31
Suspended biomass	35	4	34	8	49	12
Total COD out	726	84	292	70	227	56
Attached biomass	137	16	126	30	179	44

The SCOD removal efficiencies for butyrate and propionate were 88 ± 5.3 and 87 ± 5.8%, respectively. The relatively lower SCOD removal efficiencies for butyrate and propionate compared with acetate was consistent with the measured current densities.

The possible COD sinks in this systems were soluble microbial products (SMP), biomass (suspended and attached) and hydrogen. No methane was detected during the whole experiment. The initial TCOD was equal to the initial SCOD since the MEC were fed with synthetic solids-free substrates. The COD equivalent of hydrogen was calculated by Eq. (12), and the calculation for suspended biomass is in Eq. (13) (Lee et al., 2008). 1 mL of hydrogen is equivalent to 0.654 mg COD at room temperature (25 °C). The suspended biomass was obtained from the COD data. The SMP was determined from the soluble COD. Thus, the only unknown COD sink is the

attached biomass, which can be estimated from the COD mass balance. Without considering the COD from the attached biomass, the COD closure for acetate, butyrate and propionate-fed cycles were 84 ± 10.1, 70 ± 6.5 and 56 ± 4.7%, respectively:

$$1 \text{ mL } H_2 = \frac{1 \text{ mmol } H_2}{22.4 \text{ mL}} \frac{273.15 \text{ K}}{298.15 \text{ K}} \frac{2 \text{ meq } e^-}{\text{mmol } H_2} \frac{8 \text{ mg COD}}{\text{meq } e^-} = 0.654 \text{ mg COD} \quad (12)$$

$$\text{Suspended biomass} = (\text{TCOD} - \text{SCOD})_{\text{final}} - (\text{TCOD} - \text{SCOD})_{\text{initial}} \quad (13)$$

The results showed that the suspended biomass in the acetate-fed MEC only accounted for 4% of the initial COD. Because the acetate is well known to be readily degradable by ARB, the acetate fed MEC is more favorable for ARB growth compared to acetogenic bacteria. In the acetate-fed MEC, the lower the suspended biomass the better, since it inferred that more ARB were attached. This was consistent with the higher current density achieved in acetate-fed MEC. The calculated suspended biomass yield and the estimated attached biomass in the acetate-fed MEC were 0.042 g biomass COD/g substrate COD and 0.166 g biomass COD/g substrate COD, respectively. These data are similar to the literature. Lee et al. (2008) observed a suspended biomass yield of 0.058 g biomass COD/g acetate COD, and attached biomass yield of 0.117 g biomass COD/g acetate COD in a batch H-type MFC. However, with different substrate, the biomass growth would be highly different. For instance, the suspended and attached biomass observed for glucose by Lee et al. (2008), were 0.119 g biomass COD/g glucose COD, and 0.202 g biomass/g glucose COD, respectively, confirming that the substrate in MEC has a significant impact on the biomass yield and attachment. The biomass yields for butyrate-fed and propionate-fed MEC in this experiment are listed in Table 4.

The relatively higher biomass yields achieved in the propionate-fed and butyrate-fed MEC were contradictory with the relatively lower current densities than the acetate-fed MEC. These results emphasize that the butyrate and propionate could not be consumed by ARB directly. Instead, they were oxidized by acetogenic microorganisms first. In the propionate-fed and butyrate-fed MEC, the acetogenic microorganism became more active than in the acetate-fed MEC, and accordingly affected to the higher biomass yield. Even though the acetogenic microorganisms could oxidize propionate and butyrate, they could not transfer the electrons to the anode electrode, and therefore less current densities were observed. It is noteworthy that the ratio of

Table 5

Average biomass yield based on the COD removed for each substrate.

	Biomass yield g biomass COD/g substrate COD		
	Acetate-fed	Butyrate-fed	Propionate-fed
Suspended biomass	0.042	0.092	0.140
Attached biomass	0.166	0.341	0.510
Total	0.208	0.433	0.650

suspended biomass to attached biomass was approximately 1:4 for all the three substrates, indicating that the attachment characteristics of the various microbial groups to the anode surface were similar since the acetate-fed MEC biomass was predominantly ARB, while both the butyrate and propionate-fed MECs biomass comprised both acetogenic bacteria and ARB. Normalizing the maximum current density to the observed attached biomass is a measure of the activity of ARB and electron transfer capabilities, which as shown in Table 5, yields 43.8, 20.1 and 8.7 A/mg biomass COD/m² for the acetate-fed, butyrate-fed, and propionate-fed cycles. This reveals that potentially both the activity and electron transfer capabilities of the attached biomass in the butyrate-fed MEC was more than twice that of the propionate-fed MEC. These results further demonstrated that, compared with butyrate, propionate was more difficult to degrade in the MEC.

4. Conclusion

This study mainly focused on the comparison of different parameters in MEC feed with different VFAs (acetate, butyrate and propionate). Each substrate was fed to the reactor for three consecutive-batch cycles. Of the three operational MECs, the acetate-fed MEC exhibited the best overall performance, whereas the propionate-fed MEC achieved the worst performance, which demonstrated that propionate could not be utilized by anode respiring bacteria as easily as butyrate. The ratio of the suspended biomass to attached biomass was approximately 1:4 for all the three substrates.

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